THE UNIVERSITY OF CHICAGO

MASTERS IN COMPUTATIONAL SOCIAL SCIENCE
THE UNIVERSITY OF CHICAGO

MACS 30111



The fundamental package for scientific computing with Python

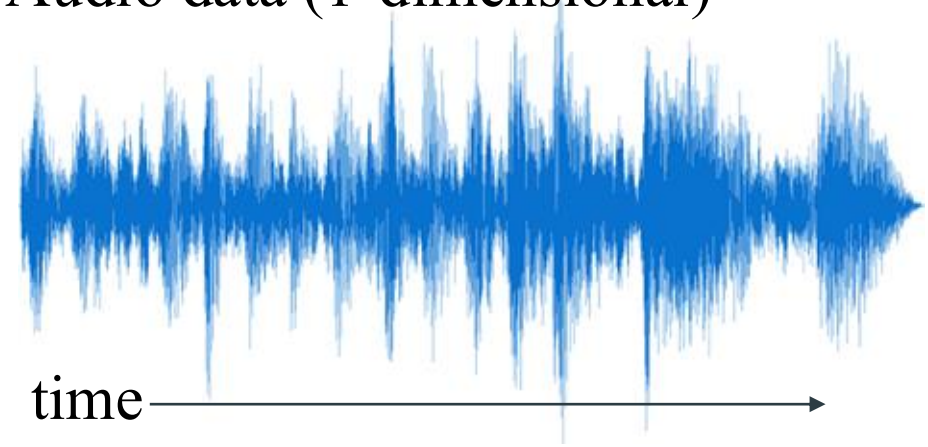
Chapter: 4.2
Documentation: https://numpy.org/devdocs/use

Topics:

- Introduction and Motivation for array programming
- Creating *Numpy* arrays
- Indexing into *Numpy* arrays
- Working with arrays and array operations
- Advanced array manipulations
- An extended example of standardizing features

The need for high-dimensional array

Audio data (1-dimensional)



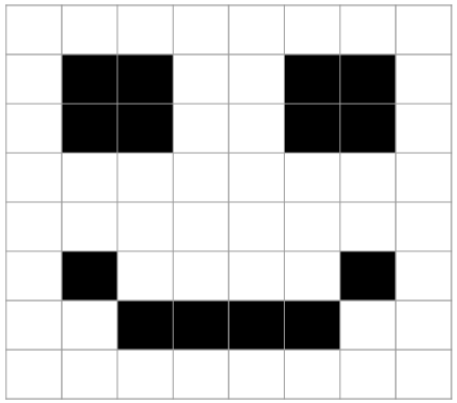

```
array([-0.00270751, -0.00303302, -0.00159557, ..., -0.0012889 ,
       -0.00184731, -0.00210062], dtype=float32)
```

Kaggle/UrbanSound8K

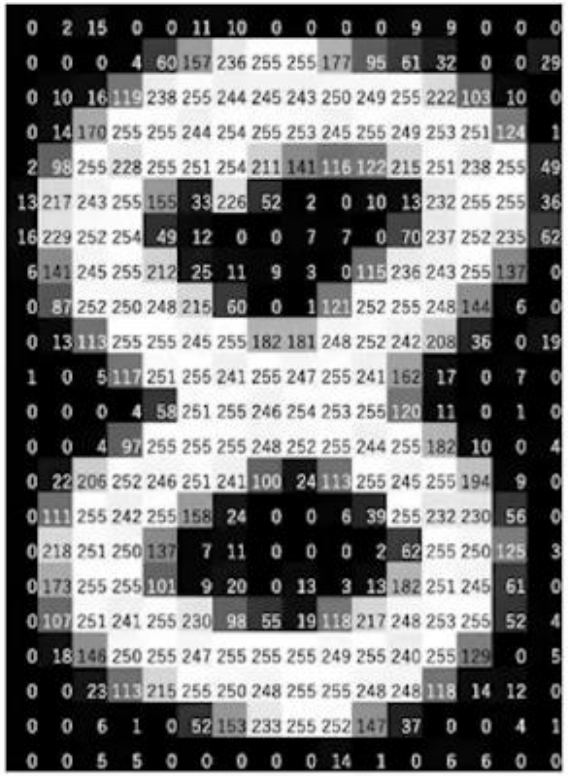
- Machine learning
- Deep learning

The need for high-dimensional array

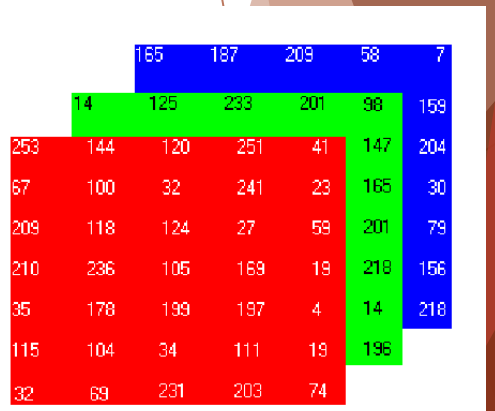
Image data



0	0	0	0	0	0	0	0
0	1	1	0	0	1	1	0
0	1	1	0	0	1	1	0
0	0	0	0	0	0	0	0
0	0	0	0	0	0	0	0
0	1	0	0	0	0	1	0
0	0	1	1	1	1	0	0
0	0	0	0	0	0	0	0







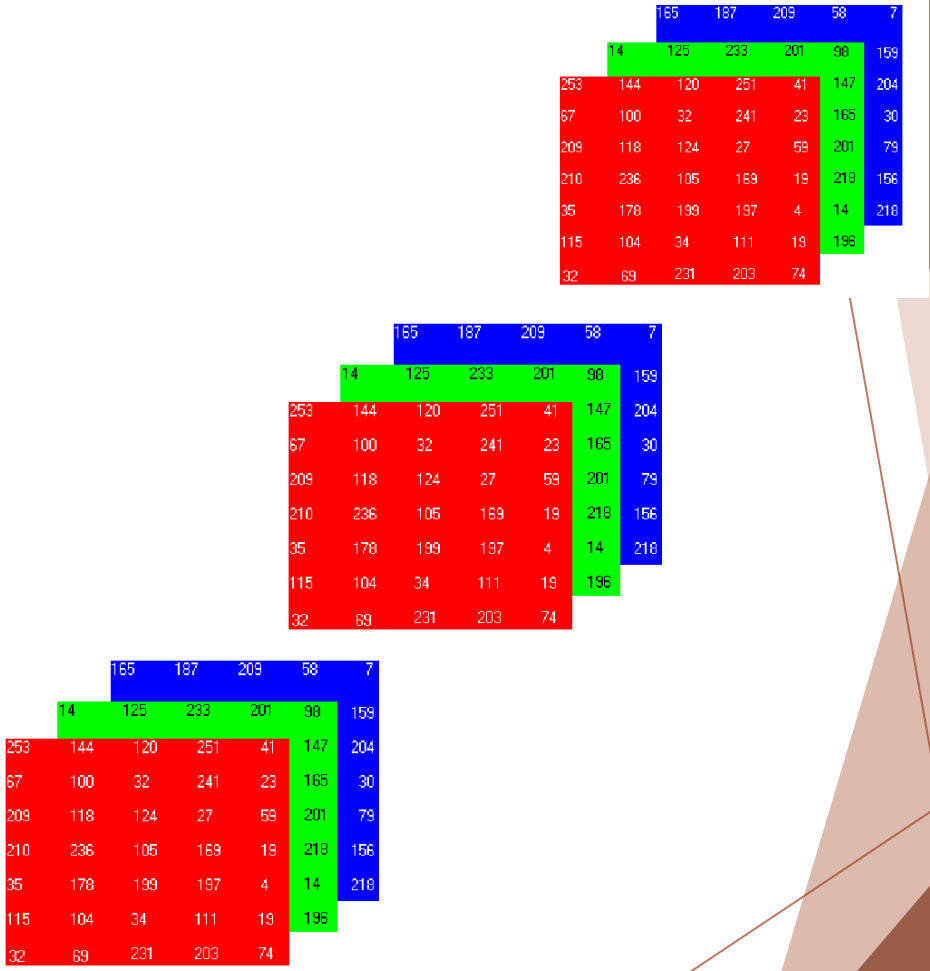
[Kaggle/Animals 151]

- Machine learning
- Deep learning

The need for high-dimensional array

Video data





[Kaggle/youtube-video-dataset]

- Machine learning
- Deep learning

The need for high-dimensional array

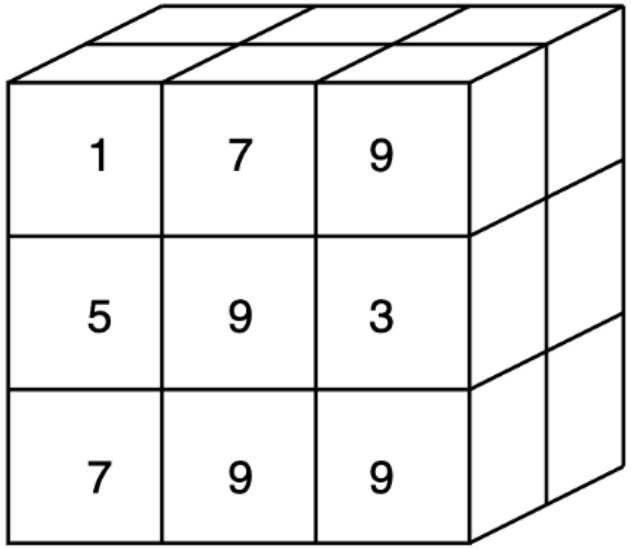
1D Array

3	2

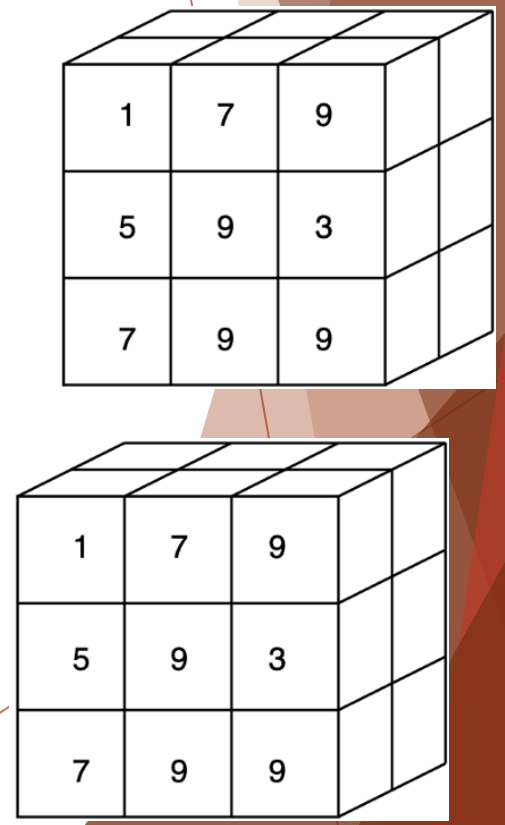
2D Array

1	0	1
3	4	1

3D Array



4D Array



Nested Lists

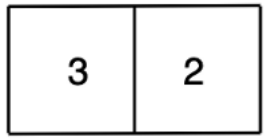
```
[["M", "M", "F", "M", "M"],
 ["F", "B", "B", "B", "F"]
 ["M", "M", "M", "M", "B"]
 ["B", "B", "B", "F", "B"]
 ["B", "M", "M", "M", "F"]]
```

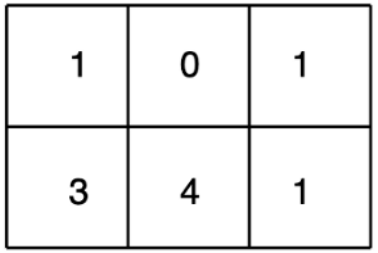
Benefits: Easy to code up
(you already know how!)

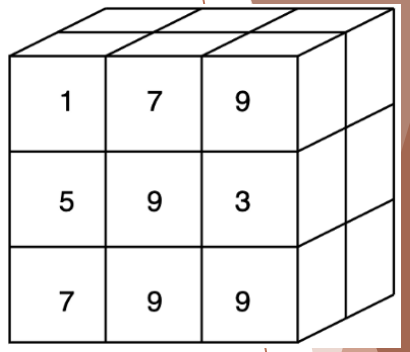
(0,0) 0	(0,1) 0	(0,2) 1	(0,3) 1	(0,4) 0
(1,0) 0	(1,1) 1	(1,2) 1	(1,3) 2	(1,4) 0
(2,0) 0	(2,1) 0	(2,2) 2	(2,3) 2	(2,4) 1
(3,0) 1	(3,1) 2	(3,2) 1	(3,3) 1	(3,4) 2
(4,0) 0	(4,1) 1	(4,2) 1	(4,3) 0	(4,4) 2

Drawbacks

Sum the elements in 1, 2, and 3-dimensional arrays:








```
sum = 0
for i in range(N):
    sum += one_dim_list[i]
```

```
sum = 0
for i in range(N):
    for j in range(M):
        sum += two_dim_list[i][j]
```

```
sum = 0
for i in range(N):
    for j in range(M):
        for k in range(N):
            sum +=
three_dim_list[i][j][k]
```

- Writing dimension-independent **generic** code is difficult when using lists-of-lists.

Drawbacks: *for* loop VS *numpy* operation

```
X = make_big_list(10000000)

total_sum = 0
for i in range(10000000):
    total_sum += X[i]
```

1.17 seconds

```
X = make_big_array(10000000)


total_sum = np.sum(X)
```

0.007 seconds

Numpy operation is faster

- Writing your own for loops can be **slow**. Performance matters when working with large datasets!

Numpy: scientific computing with n-d array



NumPy (pronounced “numb-pie”) is an open-source library in Python. It supports:

1. Multidimensional data structure (the ndarray)
2. Fast mathematical operations
3. An interface that reuses much of the Python list interface

Tradeoff: NumPy arrays can only hold values of the same *type*

Python list can hold different data types.
A NumPy array can only hold the same type: integers, floats, Boolean.

- Most numpy operations are performed on all elements together, different data types have different operations.

```
1  a = [1, 2.0, 'cat', False]
2  type(a)
```

```
list
```

```
1  import numpy as np
2  b = np.array([True, False, False])
3  type(b)
```

```
numpy.ndarray
```

Reference:

- https://numpy.org/devdocs/user/absolute_beginners.html
- https://numpy.org/doc/stable/user/basics.types.html

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Creating new NumPy arrays

- From existing lists
- Creating arrays with the same value
- Creating arrays with a range of values (e.g., *arange(), linspace()*)
- Joining arrays (e.g. np.array([array1, array2]))
- The data type, or “dtype”, of an array
- Array Shape

Coding practice: 4.2.2

Creating an array from an existing list

First, we usually **import** the NumPy library using a shorter name, "np"

```
import numpy as np
```

Then we can **create** a new array from a list

```
x = np.array([1, 2, 3])
```

We can then **index** into the array like a list:

```
1 print(x[2])
```

```
3
```

Coding practice: 4.2.2

Creating multi-dimensional arrays from nested lists

To create a two-dimensional array, we can use two nested lists:

```
y = np.array([[1, 2, 3], [4, 5, 6]])
```

To print this array:

```
1 print(y)
```

```
[[1 2 3]
 [4 5 6]]
```

To access elements in this array:

```
1 print(y[1,2])
```

```
6
```

Coding practice: 4.2.2

Creating arrays: constant values

If you need to create an array of all zeros, you can use *np.zeros*

```
1 np.zeros(10)
```

```
array([0., 0., 0., 0., 0., 0., 0., 0., 0., 0.])
```

You can create multidimensional arrays like this as well, by passing a **shape**-like tuple

```
1 np.zeros((3, 5))
```

```
array([[0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0.],
       [0., 0., 0., 0., 0.]])
```

Numpy also supports creating arrays of ones like this

```
1 np.ones((5, 8))
```

```
array([[1., 1., 1., 1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1., 1., 1., 1.],
       [1., 1., 1., 1., 1., 1., 1., 1.]])
```

Coding practice: 4.2.2

Creating arrays: ranges

Numpy supports creating arrays of numerical ranges with the ***arange*** and ***linspace*** functions.

```
np.arange(start, stop, step)
```

exclusive, just like Python's range()

step size

```
np.linspace(start, stop, num)
```

first value in array

last value in array, **inclusive**

num: how many values (not steps!)

```
1 np.arange(1,5)
```

```
array([1, 2, 3, 4])
```

```
1 np.linspace(0, 1, 5)
```

```
array([0.  , 0.25, 0.5 , 0.75, 1.  ])
```

Coding practice: 4.2.2

Reference:

- https://numpy.org/doc/stable/reference/generated/numpy.arange.html
- https://numpy.org/doc/stable/reference/generated/numpy.linspace.html#numpy.linspace

Array types

Remember that the contents of NumPy arrays must all be of the **same type**.

***.dtype*:** the array's data type

```
1  x = np.array([1, 2, 3])
2  print(x)
```

```
[1 2 3]
```

```
1  x.dtype
```

```
dtype('int64')
```

Integers (64-bit)

```
1  x = np.array([1.2, 2.5, 3.7])
2  print(x)
```

```
[1.2 2.5 3.7]
```

```
1  x.dtype
```

```
dtype('float64')
```

Floating-point (64-bit)

Coding practice: 4.2.2

Array dimensions: shape

The ***.shape*** attribute: a tuple describing the length in each dimension

```
1  x = np.array([1, 2, 3])
2  x.shape
```

```
(3,)
```

```
1  y = np.array([[1, 2, 3],
2                [4, 5, 6]])
3  y.shape
```

```
(2, 3)
```

- The length of the shape attribute : the number of dimensions
- A 1-D array is neither a row nor a column vector (x.T does nothing)

Coding practice: 4.2.2

Combining arrays

- **np.array([a, b])** adds a new dimension

```
a = np.zeros((3, 5)); b = np.ones((3, 5))
np.array([a, b]).shape       # (2, 3, 5)
```

- **np.concatenate** joins along an existing axis

```
np.concatenate([a, b]).shape          # (6, 5)
np.concatenate([a, b], axis=1).shape  # (3, 10)
```

- **Shortcuts:** np.vstack([a, b]) stacks rows; np.hstack([a, b]) stacks columns

Numpy: methods and routines

zeros: pass (*shape tuple, dtype*)

ones: pass (*shape tuple, dtype*)

rng.random: pass (**size tuple**)

arange: pass ([*start*,]*stop*, [*step*,]*dtype*)

linspace: pass (*start, stop, num=50, endpoint=True, dtype=None, axis=0*)

loadtxt: loads data into an array (needs file name)

https://numpy.org/doc/stable/reference/

Random numbers and seeds

- **A seed** makes “random” results reproducible: everyone gets the same numbers

```
rng = np.random.default_rng(0)   # seed = 0
rng.random((3, 5))               # floats in [0, 1)
rng.integers(0, 11, (5, 5))      # 0-10, not 11!
rng.standard_normal((20, 20))    # mean 0, sd 1
```

- **Older style you’ll see online:** np.random.seed(0) and np.random.randint(0, 11, (5, 5))

Task 1

- Create an array of 3 x 5 filled with zeros
- Create an array of 3 x 5 filled with ones
- Create an array of 3 x 5 that has random values from 0 to 1 in it

- Then, combine all these arrays into one array.
  - Find its shape and data type

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Agenda

- Quiz (6 qs)
- Notebook
- Final exam
- PA 7: experiement!

One-dimensional array indexing

Same as a Python list:

```
1  x = np.array([1, 2, 3])
2  print(x[2])
```

```
3
```

Coding practice: 4.2.3

Multi-dimensional array indexing

We can create a two dimensional array

```
1  y = np.array([[1, 2, 3], [4, 5, 6]])
2  y
```

```
array([[1, 2, 3],
       [4, 5, 6]])
```

What if we only use one dimension? *y[0]*

```
[[1 2 3]
 [4 5 6]]
```

And index with *y[0, 1]*

```
[[1 2 3]
 [4 5 6]]
```

two indices separated by a comma

Coding practice: 4.2.3

Array slicing

The same Python slicing operations

```
y = np.array([[1,  2,  3,  4],
              [5,  6,  7,  8],
              [9, 10, 11, 12]])
```

y[0,:3]

```
[[1  2  3  4]
 [5  6  7  8]
 [9 10 11 12]]
```

y[1:,:3]

```
[[1  2  3  4]
 [5  6  7  8]
 [9 10 11 12]]
```

Coding practice: 4.2.3

Slices are views, not copies

- **An array slice** points to the same data as the original (unlike a list slice!)

```
a = np.array([1, 2, 3, 4])
b = a[:2]
b[0] = 99
a      # array([99,  2,  3,  4])
```

- **Need an independent array?** use .copy()

```
b = a[:2].copy()
```

- **Selecting with a list or a mask** (a[[0, 2]], a[a > 2]) returns a copy

Task 2

- Set your seed to zero (why? How?)
- Create a matrix of random integers from 0 to 10 (inclusive) that is 5 x 5
- Slice your matrix so that you have the following:
  - Rows: odd numbered rows
  - Columns: numbered one through three

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Mathematical operations with arrays

- List operation: whole list as a unit
- Array operation: element-by-element

- +, -, *, /, %
- >, <, ==
- np.sin, np.cos

Array*array: not matrix operation

Coding practice: 4.2.4

Reference:

- https://numpy.org/doc/stable/reference/routines.statistics.html
- https://numpy.org/doc/stable/reference/routines.linalg.html
- https://numpy.org/doc/stable/reference/routines.logic.html

Reshaping arrays

.reshape: change the shape of an array:

- one-dimensional to two-dimensional
- two dimensional back into one-dimensional

Note:

- the new array must have the **same number of elements** as the original array
- element updates on reshaped array mirrors back to the original array

Coding practice: 4.2.5

Task 2.1

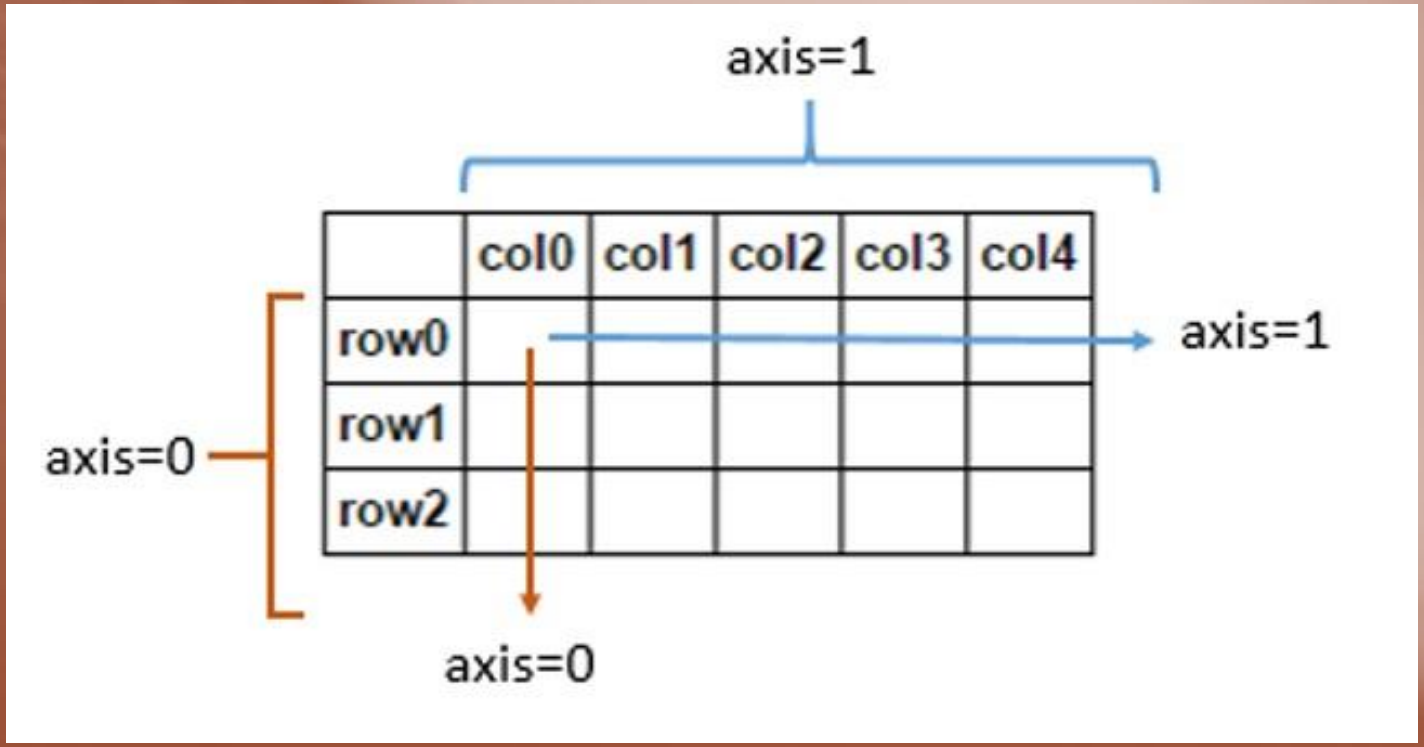
- Create a matrix that goes from 0 to 10 that has the dimensions of 3x5. Divide the range evenly across the 15 values.

Arrays and axes

Reductions along an axis

axis=0: collapse the **rows** → one result per column

axis=1: collapse the **columns** → one result per row



Reductions on arrays

Reduce arrays to lower dimensions:

- operation across all elements or along specific dimensions

```
array([[1, 2, 3],
       [4, 5, 6]])
```

```
np.sum(x, axis=0)
```

```
x[:,0], x[:,1], x[:2]
```

```
array([5, 7, 9])
```

```
array([[1, 2, 3],
       [4, 5, 6]])
```

```
np.sum(x, axis=1)
```

```
x[0,:], x[1,:]
```

```
array([ 6, 15])
```

- slicing in the axis dimension
- every index in the non-axis dimension

Coding practice: 4.2.6

Vectors, Matrices, Tensors

- **Vector** (akin to list) : one-d arrays (only one axis)
  - MATH WORKS WEIRD HERE: no row vs. column, so x.T does nothing
  - Can reshape into a ‘matrix’ (meaning you give it a second dimension)
- **Matrix** (like regular 2d matrix): two axes (0 and 1)
  - Math is OK
- **Tensor** (multiple / higher dimensions) (AAHHH!)
  - Harder to visualize but possible (good for dimension reduction!)

Fun: Images

- Import the image (Pillow library)
- Can get image info: pixels
  WEBP
  (2048, 2048)
  RGB
- Dimensions:

<class 'numpy.ndarray'>
(2048, 2048, 3)

- Red dimension: numpydata[:,:,0]

```
[[65 51 64 ... 53 61 62]
 [54 36 44 ... 47 48 57]
 [57 48 50 ... 49 44 44]
 ...
 [48 44 47 ... 40 51 52]
 [54 57 53 ... 44 58 46]
 [56 63 55 ... 71 66 53]]
```



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Advanced array manipulations

1. Fancy indexing:
   - Indexing with a list/an arrays of indexes (array indexing)
   - Indexing with an array of Booleans to select elements (masked indexing)
2. Performing operations between arrays of different dimensions ("broadcasting")

Coding practice: 4.2.7

Index a subset of elements

```
a = np.array([1, 4, 9, 16, 25, 36, 49])
```

Array indexing with:

- **One index**: `a[3]` → 16

```
a = np.array([1, 4, 9, 16, 25, 36, 49])
```

- **A list of indexes**: `a[[3, 1, 6]]` → `array([16, 4, 49])`

```
a = np.array([1, 4, 9, 16, 25, 36, 49])
```

```
array([16,  4, 49])
```

- **An array of indexes**: `a[np.array([[1, 3], [5, 2]])]`

```
a = np.array([1, 4, 9, 16, 25, 36, 49])
```

```
array([[ 4, 16],
       [36,  9]])
```

Coding practice: 4.2.7

Masked indexing

Select a subset of elements via a boolean array:

```
c = np.array([100, 200, 300])

mask = np.array([True, False, True])
```

Relational operations:

```
1 c>100
```

```
array([False,  True,  True])
```

```
1 c[c>100]
```

```
array([200, 300])
```

```
1 c[(c>100) & (c/100==2)]
```

```
array([200])
```

```
1 c[(c>100) | (c/100==2)]
```

```
array([200, 300])
```

Coding practice: 4.2.7

Task 3: bringing it all together

- **Create a 10 x 10 matrix** of random numbers between 0 and 100 (inclusive) with a seed of 10.
- **Create a sub-matrix** of this where you select even-numbered rows and odd-numbered columns. **Then: set** values ≤ 50 to zero (keep values > 50) and sum each column.

Advanced topics

- Broadcasting
- Standardizing

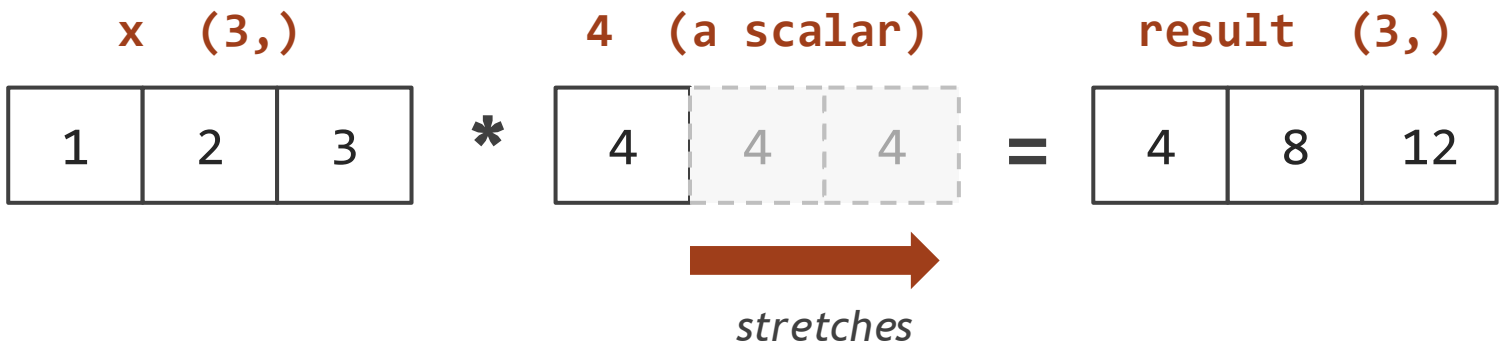
Broadcasting

- NumPy can combine arrays with **different but compatible shapes**.
- It **stretches** dimensions of size 1 to match, without copying any data, and then works element by element.

Broadcasting: a scalar stretches

A single number stretches to match every element.

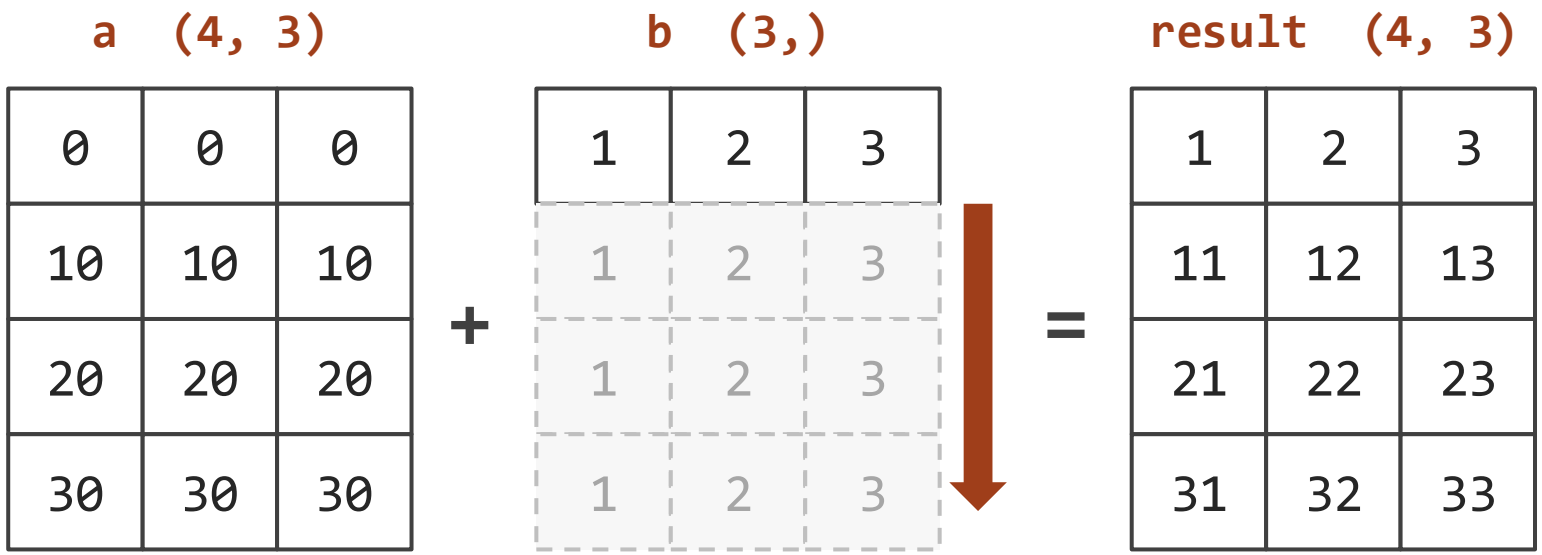
```
x = np.array([1, 2, 3])
x * 4      # array([ 4,  8, 12])
```



Gray cells are **not copied** in memory; NumPy just acts as if they were there.

Broadcasting: a row stretches down

A row `(3,)` is reused for every row of a `(4, 3)` array.



Shapes: `(4, 3) + (3,)` → `(4, 3)` Each row of a gets [1, 2, 3] added.

The broadcasting rule

Check the shapes before you run the code.

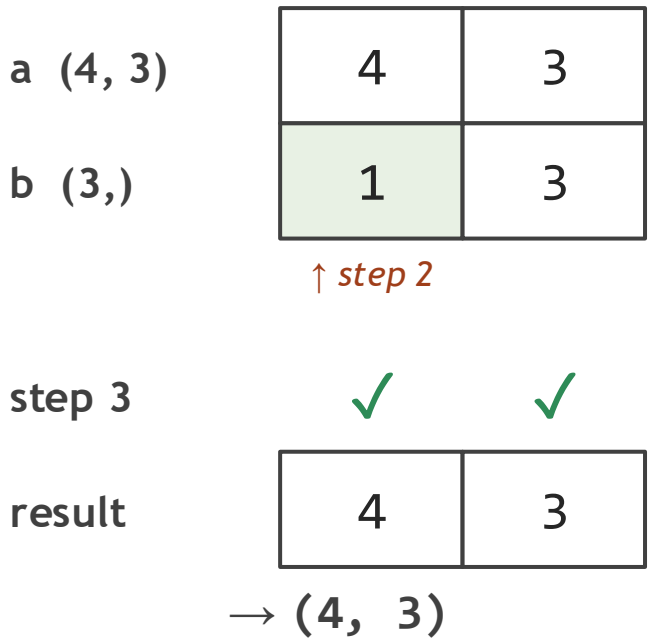
1 Line the shapes up on the **right**.

2 A missing dimension on the left counts as **1**.

3 Each column must be **equal**, or one of them **1**.

The result takes the bigger number.

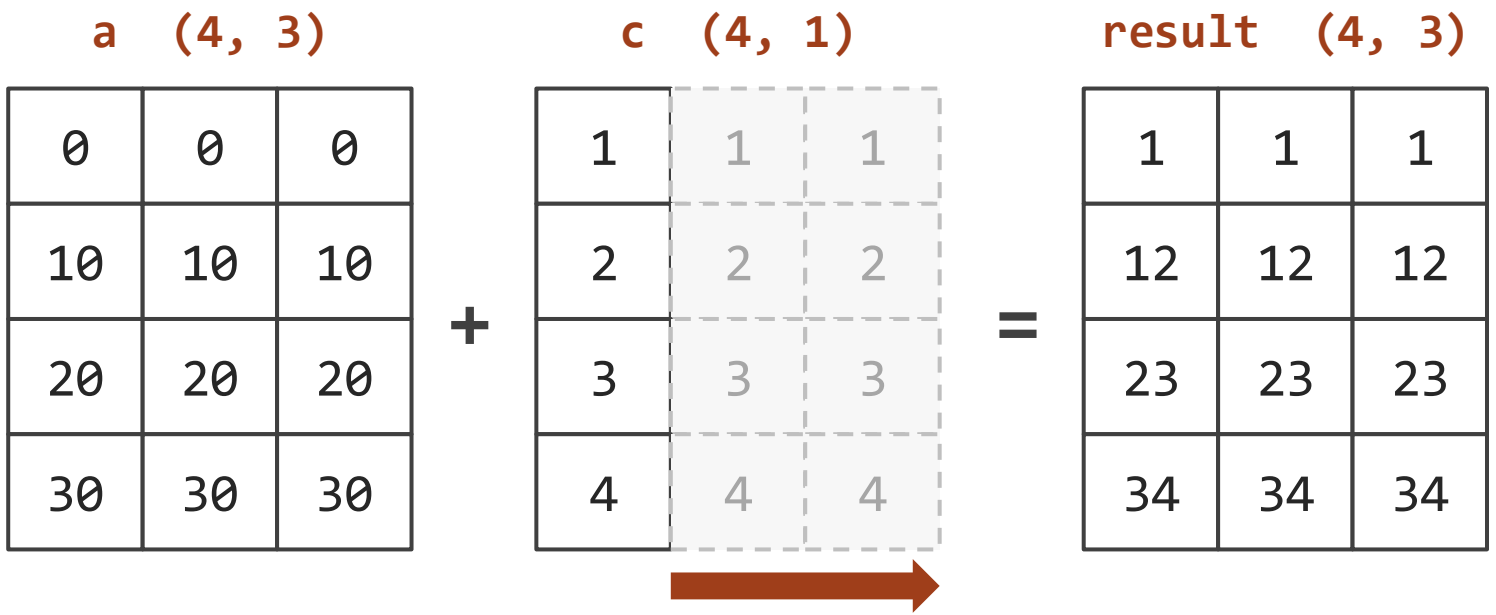
Neither? `ValueError`.



Example: `(4, 3) + (3,)`. Step 2 turns `(3,)` into `(1, 3)`: one row, which stretches down.

Broadcasting: a column stretches across

A column `(4, 1)` is reused for every column.

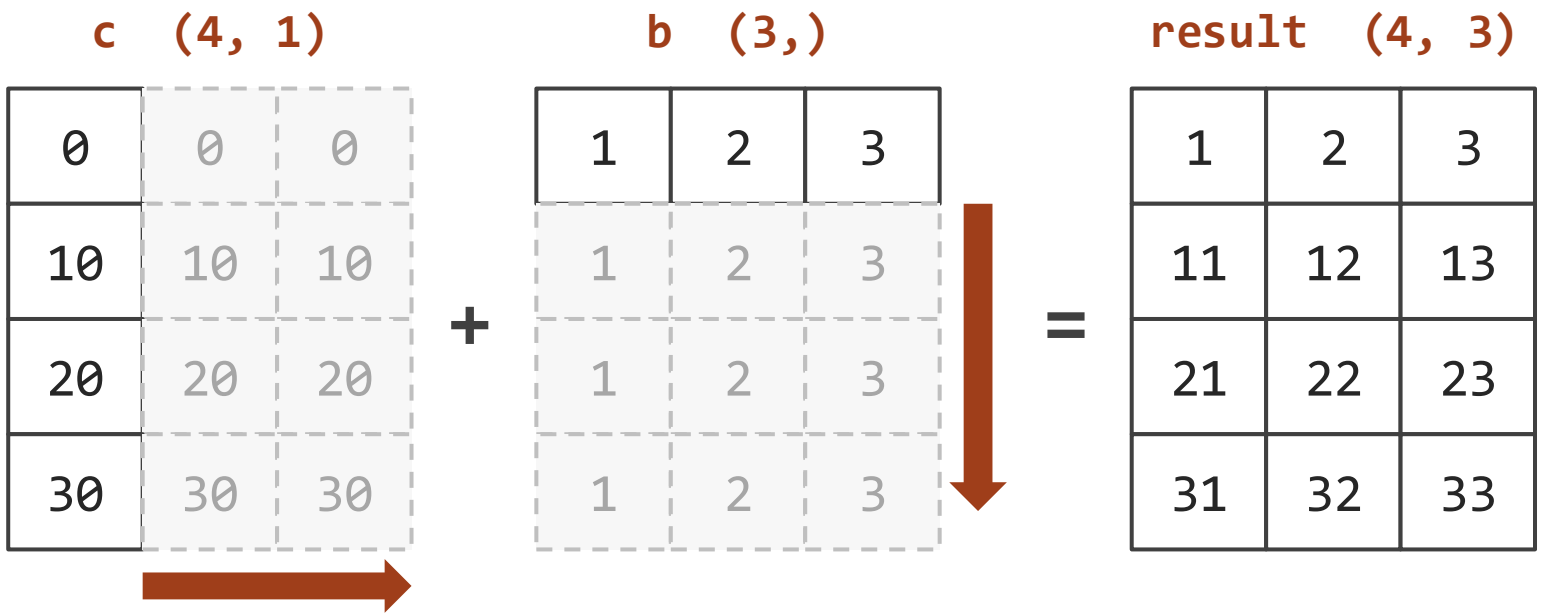


Shapes: `(4, 3)` + `(4, 1)` → 3 vs 1 ✓, 4 vs 4 ✓ → `(4, 3)`

Make a column with `c.reshape(4, 1)` or `a.mean(axis=1, keepdims=True)`.

Broadcasting: both stretch

When each array has a 1 in a different place, **both** stretch.



Shapes: `(4, 1) + (3,)` → `(4, 1) + (1, 3)` → `(4, 3)`

Quiz Q6: `Rooibos + Mate` is `(2, 1) + (3,)` → `(2, 3)`. It works!

When broadcasting fails, and the fix

Shapes that don't line up raise an error. Fix them by reshaping.

`(4, 3) + (4,)`

a	4	3
b	1	4
	✓	✗

3 vs 4: neither is 1

```
ValueError: operands could not be
broadcast together with shapes (4,3) (4,)
```



`(4, 3) + b.reshape(4, 1)`

a	4	3
b	4	1
	✓	✓

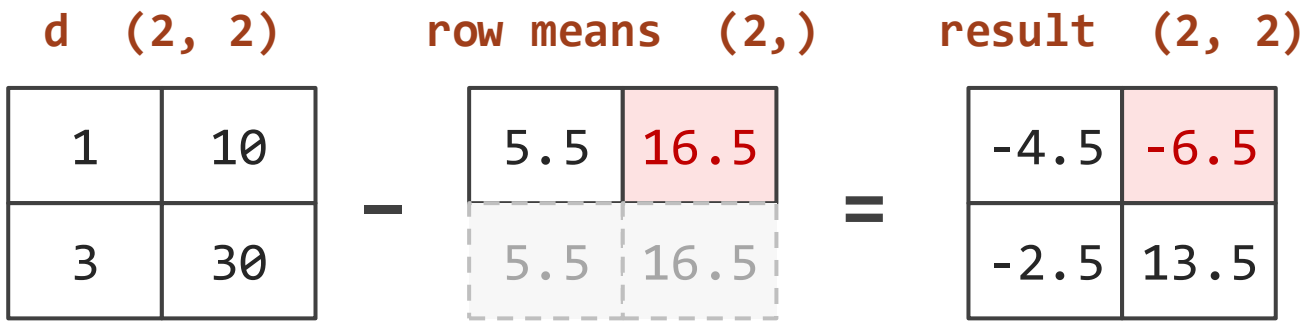
3 vs 1 ✓, 4 vs 4 ✓ → `(4, 3)`

Same as the column slide.

Mean of each row: `a.mean(axis=1)` is `(4,)`, so `a - a.mean(axis=1)` fails. Use `keepdims=True`.

Matching shapes ≠ correct

Broadcasting only checks shapes. It can't tell rows from columns.



row means:
`5.5 (row 0)`
`16.5 (row 1)`

laid across the top, then down

10 − 16.5: row 1's mean, in row 0

`(2, 2) - (2,)`: 2 vs 2 ✓ → **runs, no warning, wrong answer.**

On `(10, 4)` data this bug crashes. On **square** data it doesn't.

Check: works or error?

Use the rule. If it works, what is the result shape?

1. `(3,)  * (2,)` works → _____ or error
2. `(5, 2)  + (2,)` works → _____ or error
3. `(5, 2)  - (5,)` works → _____ or error
4. `(2, 1)  + (3,)` works → _____ or error

Hint: (5, 2) is the patients array: 5 patients, 2 measurements.

Example: broadcasting → standardizing

`patients`: height (in) and weight (lbs) for 5 patients, shape `(5, 2)`. Same data as slide 55.

Predict the shape (or the error) before you run each line.

Part A: broadcasting warm-up

1. `patients + 10` shape?
2. `patients * np.array([2.54, 0.4536])` to metric: does it work?
3. `patients - patients.mean(axis=1)` runs? If not, why, and how would you fix it?

Rule (slide 50): right to left, equal or one is 1

Example: standardize by hand

Part B: standardize the patients by column

4. `mu = patients.mean(axis=0)` and `sigma = patients.std(axis=0)`: shapes?
5. `z = (patients - mu) / sigma`: check each column has mean 0, std 1
6. Standardize the **metric** version too. Same z-scores? (`np.allclose`) Why?
7. Boolean mask: which patients are above average on **both** height and weight?

Next: this is exactly `standardize_features` (slide 61) → Standardizing task

Topics:

- Introduction and Motivation for array programming
- Creating *Numpy* arrays
- Indexing into *Numpy* arrays
- Working with arrays and array operations
- Advanced array manipulations
- **An extended example of standardizing features**

Standardizing

- Allows you to compare items with others, even when they have different scales

The need for standardized features

Patient number	Height (inches)	Weight (lbs)
11471	72	190
9412	60	124
12911	65	170
42311	75	225
32141	50	85

Patient number	Height (mi)	Weight (g)
11471	0.001136	86182.6
9412	0.0009470	56245.5
12911	0.0010259	77110.7
42311	0.0011837	102058
32141	0.0007891	38555.4

The need for standardized features

We can take an unknown feature and "**standardize**" it to have **a mean of zero and a standard deviation of one**. This way we can have some intuition about how far away a value is from "average" (zero mean).

This is a very common operation for most machine learning and statistics tasks, where often you have more than just two features -- sometimes you have two million! How would you ever make sense of all those units?

How to standardize data

```
data = [[89.0, 66.0, 23.0, 94.0],
        [137.0, 40.0, 35.0, 168.0],
        [78.0, 50.0, 32.0, 88.0],
        [197.0, 70.0, 45.0, 543.0],
        [189.0, 60.0, 23.0, 846.0],
        [166.0, 72.0, 19.0, 175.0],
        [118.0, 84.0, 47.0, 230.0],
        [103.0, 30.0, 38.0, 83.0],
        [115.0, 70.0, 30.0, 96.0],
        [126.0, 88.0, 41.0, 235.0]]
```

$$\mu_j = 1/N * \sum_{i=0}^{N-1} m_{i,j}$$

$$\sigma_j = \sqrt{1/N * \sum_{i=0}^{N-1} (m_{i,j} - \mu_j)^2}$$

$$m'_{i,j} = (m_{i,j} - \mu_j)/\sigma_j$$

The regular python way of standardizing features

```
def standardize_features(data):
    N = len(data)
    M = len(data[0])

    # initialize the result w/ NxM list of lists of zeros.
    rv = []
    for i in range(N):
        rv.append([0] * M)

    # for each feature
    for j in range(M):
        mu = compute_feature_mean(data, j)
        sigma = compute_feature_stdev(data, j)

        # standardized feature
        for i in range(N):
            rv[i][j] = (data[i][j] - mu)/sigma
    return rv
```

$$\mu_j = 1/N * \sum_{i=0}^{N-1} m_{i,j}$$

```
def compute_feature_mean(data, j):
    '''
    Compute the mean of feature (column) j

    Inputs:
      data (list of list of floats)

    Returns (float): mean of feature j
    '''

    N = len(data)
    total = 0
    for i in range(N):
        total += data[i][j]
    return total/N
```

$$\sigma_j = \sqrt{1/N * \sum_{i=0}^{N-1} (m_{i,j} - \mu_j)^2}$$

```
def compute_feature_stdev(data, j):
    '''
    Compute the standard deviation of feature (column) j

    Inputs:
      data (list of lists of floats)

    Returns (float): standard deviation of feature j
    '''

    N = len(data)
    mu = compute_feature_mean(data, j)
    total = 0
    for i in range(N):
        total = total + (data[i][j] - mu) ** 2
    return math.sqrt(1 / N * total)
```

Feature standardization with NumPy

```
def standardize_features(data):
    '''
    Standardize features to have mean 0.0 and standard deviation 1.0.
     Inputs:
       data (2D NumPy array): data to be standardized
      Returns (2D NumPy array): standardized data
    '''

    mu_vec = data.mean(axis=0)
    sigma_vec = data.std(axis=0)

    return (data - mu_vec)/ sigma_vec
```

Note: NumPy's .std() divides by N (matching the formula). pandas' .std() divides by N−1, so the numbers differ slightly. Use ddof=1 to match.

Simpler standardization with NumPy

We significantly simplify the above code with NumPy:

1. The NumPy code is less work to write
2. The functions in NumPy are already thoroughly debugged, reducing the likelihood of a mistake
3. The NumPy code will be faster, and will therefore let us handle more data.

Worksheet Party

Midterm review

Q15: lang.append returns NONE

5. Which line changes `lang` to `["Python", "R", "Go"]`?

```
1  lang = ["Python", "R"]
```

A `lang.add("Go")`

B `lang[2] = "Go"`

C `lang += ["Go"]`

D `lang.insert("Go")`

E `lang = lang.append("Go")`

Q12: how many entries per state (sets)

12. What is printed by the following code?

```
1  complaints = [{"Company": "Acme", "State": "IL"},
2                {"Company": "Bolt", "State": "NY"},
3                {"Company": "Acme", "State": "IL"},
4                {"Company": "Acme", "State": "CA"}]
5  states = {}
6  for c in complaints:
7      if c["Company"] not in states:
8          states[c["Company"]] = set()
9      states[c["Company"]].add(c["State"])
10 print({k: len(v) for k, v in states.items()})
```

A An error is raised

B `{'Acme': 3, 'Bolt': 1}`

C `{'Acme': {'IL', 'CA'}, 'Bolt': {'NY'}}`

D `{'Acme': 2, 'Bolt': 1}`

Q 15 local vars vs global vars [add_score is local, add_all is global]

15. What is printed by the following code?

```
1  total = 0
2
3  def add_score(score):
4      total = score
5      return total
6
7  def add_all(scores):
8      global total
9      for s in scores:
10         total = total + add_score(s)
11
12 add_all([3, 4])
13 print(total, add_score(10), total)
```

A `0 10 0`

B `7 10 10`

C An error is raised

D `7 10 7`

Add_all prints nothing but modifies global to 7

Print total (modified global (7))

add_score (local, so gives you 10)

Total (global, unchanged, 7)

Q16: tricky and multiple paths. Can be off a little on this one. Think pythonically

Start with easy parts: def and return.
Why not *with open*? Bc it says it needs to read a file in function

with open next, then f to read, then counts

which two are out?
* counts = len (why??)
* A vs F: A provides backup if empty

Part B · Put the lines in order 7.5 points

16. The function `count_by_zip(filename)` reads a file where each line is `name,zip`. It returns a dictionary that maps each ZIP code to how many lines have it. For example:

If the file contains

```
donors.txt
ana,60637
ben,60615
cy,60637
```

→ then

```
count_by_zip("donors.txt")
# returns {'60637': 2, '60615': 1}
```

The lines below are scrambled and their indentation has been removed. **Two lines do not belong.**

```
A.  counts[zip_code] = counts.get(zip_code, 0) + 1
B.  def count_by_zip(filename):
C.  counts[zip_code] = len([zip_code])
D.  return counts
E.  with open(filename) as f:
F.  counts[zip_code] += 1
G.  counts = {}
H.  name, zip_code = line.strip().split(",")
I.  for line in f:
```

Space to rewrite the code if you'd like (not graded)

```
1
2
3
4
5
6
7
```

(a) (3.5 points) Write the letters of the lines in the correct order.

(b) (2 points) Under each letter, write how far that line is indented: **0, 1, 2 or 3** levels (0 = not indented).

	1	2	3	4	5	6	7
Letter							
Indent							

Correct ans

```
def count_by_zip(filename):
    counts = {}
    with open(filename) as f:
        for line in f:
            name, zip_code = line.strip().split(",")
            counts[zip_code] = counts.get(zip_code, 0) + 1
    return counts

count_by_zip("donors.txt")
```

✓ 0.0s

```
{'60637': 2, '60615': 1}
```

Additional information and practice

Quiz

Assume *x = np.array([1.0, 2.0, 3.0])*.
Which is closest to the output of *x / x.sum()*

A. 2
B. np.array([0.166, 0.333, 0.5])
C. np.array([1.0, 2.0, 3.0])

Additional fun!

- np.where:
  - Can use within a matrix to select indices based on conditions
  - Can use as a standalone for conditional replacement

```
m = np.array([[3,2,1],[3,2,1],[1,2,3]])
m[np.where(m>1)]
# → [3 2 3 2 2 3]  (same as m[m>1])
np.where(m>1, m, m+2)
# → [[3 2 3] [3 2 3] [3 2 3]]
```

List VS array

- Different data types allowed
- +: concatenation
- Pop(), append(), extend()
- List as a whole unit

- Efficiency
- Index[], [], []

- Same data type required
- Mathematical
- Element-wise
- Broadcasting
- Fast, generic
- Index, [, , ,]
- reshape

NumPy: methods and routines

- zeros: pass (***shape tuple, dtype***)
- ones: pass (***shape tuple, dtype***)
- rng.random: pass (**size tuple**)

- arange: pass ([***start***,]***stop***, [***step***,]***dtype***)
- linspace: pass (***start, stop, num=50, endpoint=True, dtype=None, axis=0***)
- loadtxt: loads data into an array (needs file name)
- savetxt: saves an array to a file (needs file name)
- where: can return indices or conditional replacement

https://numpy.org/doc/stable/reference/

(On your own!) Task 4

- Create a 20 x 20 array filled with random numbers drawn from a standard normal distribution use seed = 13
- Export it to a .csv file (np.savetxt)
- Sum along rows (one total per row → shape (20,))
- Sum along columns (one total per column)
- Calculate the overall average

Quiz

X is an array with X.shape = (5, 10). How many elements are returned by X[:3, :2]?

- 3
- 5
- 6

Quiz

x = np.array([1, 2, 3, 4]). What is the value of x * 2?

- A. np.array([1, 2, 3, 4, 1, 2, 3, 4])
- B. np.array([3, 4, 5, 6])
- C. np.array([2, 4, 6, 8])

Quiz

Consider the NumPy arrays x = np.array([1, 3, 5, 7, 9]) and y = np.array([1, 2, 3, 2, 1]).
What is the value of the expression x [y < 2] ?

a. np.array([True, False, False, False, True])
b. np.array([True, 3, 5, 7, True])
c. np.array([1, 9])
d. np.array([1, 3, 7, 9])